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Development of functionalized MWCNT-Nafion modified glassy carbon electrode as a sensor for electrochemical studies and voltammetric determination of melphalan, based on improved kinetic parameters

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Abstract

A simple, fast, sensitive and validated voltammetric method was developed to study redox behavior, electrode kinetics, redox mechanism and for determination of melphalan. Cyclic voltammetry (CV) and differential pulse voltammetry (DPV) techniques were performed for voltammetric studies of melphalan using functionalized multiwalled carbon nanotubes-Nafion modified glassy carbon electrode (f-MWCNT-Nafion/GCE). Obtained results were compared with bare glassy carbon electrode (GCE). A sharp oxidation peak was obtained in anodic direction at a potential of 0.821 V versus Ag/AgCl in Britton-Robinson (BR) buffer of pH 7.2. Absence of peak in reverse scan suggested the irreversible nature of oxidation process. Rise in diffusion coefficient and heterogeneous rate constant prove the catalytic activity of f-MWCNT. Furthermore, under optimized conditions peak current linearly increased from 1.7×10^{-8} M to 3.9×10^{-7} M concentration of melphalan with the limit of detection (LOD) and limit of quantification (LOQ) 1.95×10^{-9} and 1.3×10^{-8} respectively. The developed method was successfully applied for the determination of drug in bulk, pharmaceutical formulation and in serum sample.

Keywords: Melphalan, functionalized multiwalled carbon nanotubes, voltammetry, LOD, LOQ

1. Introduction

Melphalan, chemically 4-{bis (2-chloro ethyl) amino}-L-phenyl alanine (Figure 1), is a phenylalanine derivative of mechlorethamine and belongs to the class of nitrogen mustard^[1]. It is an antineoplastic and alkylating agent. It is used mainly in the treatment of multiple myeloma and carcinoma of breast and ovary^[2, 3]. An alkylating agent inserts an alkyl group (C_nH_{2n-1}) to DNA. Melphalan have the bis (2-chloroethyl) amine group, which forms aziridinium by intramolecular displacement of chloride by the nitrogen. The aziridinium group nonspecifically alkylates at N⁷ position of the guanine bases to form interstrand crosslinks in Deoxyribonucleic Acid (DNA) and thus blocks DNA replication and transcription which leads to the death of cancerous cell^[4, 5].

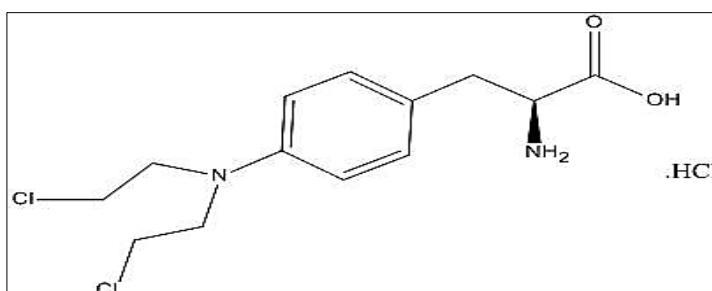


Fig 1: Chemical structure of melphalan hydrochloride

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In literature, few analytical methods have been reported viz. QCM (quartz crystal microbalance) sensing of melphalan via electropolymerized molecularly imprinted polythiophene films^[6] ion-pair high performance liquid chromatography,^[7] reverse phase high performance liquid chromatography,^[8] liquid chromatography,^[9] thin layer chromatography,^[10] gas chromatography-mass spectrometry^[11] etc. These methods are tedious, relatively expensive and involve time consuming separation and extraction processes prior to the determination of drug. In literature only one research paper was found which deals with differential pulse voltammetric determination of melphalan at glassy carbon disk electrode^[12]. But, as per the extensive literature survey, there is no voltammetric study of melphalan on functionalized multiwalled carbon nanotubes-Nafion modified glassy carbon electrode (f-MWCNT-Nafion/GCE). So, in present paper differential pulse voltammetric (DPV) technique, based on f-MWCNT-Nafion/GCE, is developed as a fast, sensitive, selective, low cost and validated technique for determination of melphalan in bulk form, pharmaceutical form and serum as biological sample. This paper also deals with the detailed electrochemical study, electrode kinetics and oxidation mechanism of melphalan under optimized pH and buffer conditions.

2. Experimental

2.1 Apparatus

All voltammograms were recorded using a model Dy 2000 series multi-channel potentiostat (Digi-Ivy, Inc., P.O. Box 200334, Austin, TX 78720-0334, USA). Chronocoulometry was performed on model 1230 A [SR 400] (CHI instrument, Bee Cave, TX, USA) electrochemical analyzer. Controlled potential coulometric experiments were carried out using the model 760 electrochemical workstation (CHI instrument). A three electrode cell assembly (fitted with glassy carbon as a working electrode, a platinum wire as an auxiliary electrode and Ag/AgCl as reference electrode) was employed for all voltammetric measurements. A calibrated CHINO digital pH meter, equipped with a glass electrode, was used for all pH-metric measurements. All experiments were performed at a standard temperature of 25°C. XRD (X-ray diffraction) studies were carried out using XRD X'Pert PRO instrument model DY2006 of PANalytical.

2.2 Materials

Melphalan hydrochloride (99% purity) was obtained as a gift from Emcure Pharmaceuticals Ltd. and used without further purification. A stock solution of 2.5×10^{-3} M of melphalan (bulk) was prepared by dissolving its precisely weighed amount in double distilled ethanol. Britton-Robinson (BR) buffer was prepared and pH adjustments were carried out using 1 M HCl and 1 M NaOH. KCl solution (1 M) was used as supporting electrolyte. MEGVAL-50mg vials containing 50 mg melphalan hydrochloride were purchased from commercial sources for pharmaceutical analysis. All chemicals were purchased from Sigma-Aldrich and were of analytical reagent grade. Double distilled deionized water, obtained from laboratory distillation assembly, was used for preparing solutions and for washings throughout the experiment. All solutions were protected from sunlight, stored at 4°C in a refrigerator and used within 24 hours to avoid decomposition. However, electrochemical responses did not show any significant changes even after 72 hours.

2.3 Methods

2.3.1 Pharmaceutical sample preparation

A quantity of 17.1 mg of melphalan hydrochloride was accurately weighed and dissolved in 20 mL of ethanol followed by sonication for 2 minutes to get homogenous standard stock solution of 2.5×10^{-3} M concentration. To record voltammetric measurements, working solutions were prepared by diluting aliquot (appropriate volume) of the standard stock solution of drug with BR buffer and 1 M KCl solution in the voltammetric cell. Effective concentration (E.C.) of the solution in the electrochemical cell was calculated using the formula $(E.C. = \text{Conc. (M) of stock drug solution} \times \text{Volume (ml) of stock drug solution added in the cell} / \text{Total volume (ml) of solution in the cell})$. Throughout research paper concentration terms were expressed in terms of E.C.

2.3.2 Spiked serum sample preparation

Drug-free human blood sample, collected from a healthy volunteer after written consent, was centrifuged in Labspin TC 450 D centrifuge machine at 4000 rpm for 30 minutes at room temperature and stored frozen until assay. The supernatant (serum) was collected and treated with acetonitrile (in the ratio of 1.2 ml acetonitrile for 2 ml serum sample) as serum denaturalizing and precipitating agent. After vortexing for 1 minute, serum sample was centrifuged for 10 minutes at 5000 rpm to eliminate serum protein residues followed by fortification with melphalan hydrochloride in ethanol to get final concentration of 2.5×10^{-3} M. Now appropriate volumes of this spiked serum sample were transferred into electrochemical cell containing BR buffer of pH 7.2 and 1 M KCl solution to get final concentration of melphalan hydrochloride as 30 µg/L (8.8×10^{-8} M) for analysis. The analyzed concentration of melphalan hydrochloride laid within the linearity range of the developed technique.

2.3.3 Pretreatment of Glassy Carbon Electrode (GCE)

To furnish reproducible electrode surface and to improve sensitivity and resolution of voltammetric peaks GCE was mechanically polished, just before each voltammetric measurement, with aqueous slurry of alumina powder (0.08 µm). Then GCE surface was washed with water followed by sonication for 45 seconds to remove any alumina particle if remained on its surface. Now, GCE was dried in an oven at 40°C for 2 minutes. Before each voltammetric measurement, a continuous stream of nitrogen gas (Instrumental Grade; 99.999% pure) was passed through solution in electrochemical cell for 10 minutes to get oxygen free environment.

2.3.4 Preparation of f-MWCNT suspension and XRD spectrum of functionalized MWCNTs

An amount of 4 mg of inactive MWCNTs was acidified with 4 mL of acid solution (HNO_3 and H_2SO_4 in 1:3 respectively) by sonication for 2 hr. Now, acidified MWCNTs were washed with double distilled water, until water gave pH 7.0, and allowed to dry in incubator. This chemical attack caused rupture of CNTs' wall and generation of functional groups due to oxidation. This was verified by XRD spectra of both inactive and activated CNTs. The radiation used was Cu-K α 2 with a wavelength of 0.1544nm to provide the estimate of atomic spacings in crystalline substances based on Bragg's law. The diffraction angle (2 θ) ranged from 0 to 30°.

XRD study revealed that inactive and activated CNTs both showed one peak at around diffraction angle of 26° corresponding to its graphitized arrangement of carbon atoms. The 2θ of 26° could be indexed as the reflection from (002) Miller plane with an interplanar spacing (d) of approximate 0.348 nm which was lesser than 0.376 nm (interplanar spacing for graphite). This could be attributed to the curvature present in the walls of MWCNTs^[13]. The acid attack caused rupture of walls of MWCNTs and generation of oxidized products at its surface. All prominent peaks from the reflection planes in the oxidized form of MWCNTs were obtained as (Miller plane; (001), $2\theta = 9.37^\circ$, $d = 0.943$ nm), (Miller plane; (101), $2\theta = 12.65^\circ$, $d = 0.699$ nm), (Miller plane; (021), $2\theta = 19.45^\circ$, $d = 0.456$ nm), (Miller plane; (202), $2\theta = 23.02^\circ$, $d = 0.36$ nm) (Miller plane; (002), $2\theta = 25.22^\circ$, $d = 0.353$ nm), (Miller plane; (006), $2\theta = 28.96^\circ$, $d = 0.308$ nm) and (Miller plane; (222), $2\theta = 31.78^\circ$, $d = 0.28$ nm (powder diffraction file2 00-045-1940).

Moreover, interplanar spacing of 0.353 nm {plane (002) and $2\theta = 25.22^\circ$ } corresponding to the graphitized structure in f-MWCNTs was higher than that of unfunctionalized one (0.348 nm) and diffraction angle shifted from 26° to 25.22° with a stronger peak than that of unfunctionalized MWCNTs. The reason for these changes could be attributed to the introduction of surface oxide products into the opened tips of f-MWCNTs causing an expansion in its interplanar spacing while acid attack caused shifting in diffraction angle by increasing its degree of graphitization via transformation of amorphous carbon respectively (Figure 2)^[14, 16]. Hence, XRD analysis well differentiated functionalized MWCNTs from unfunctionalized one. Literature supports the generation of -COOH as functional group upon oxidation of MWCNTs by means of acid mixture attack^[16]. Hence -COOH group was considered as prime functional group to be present at the modified electrode's surface.

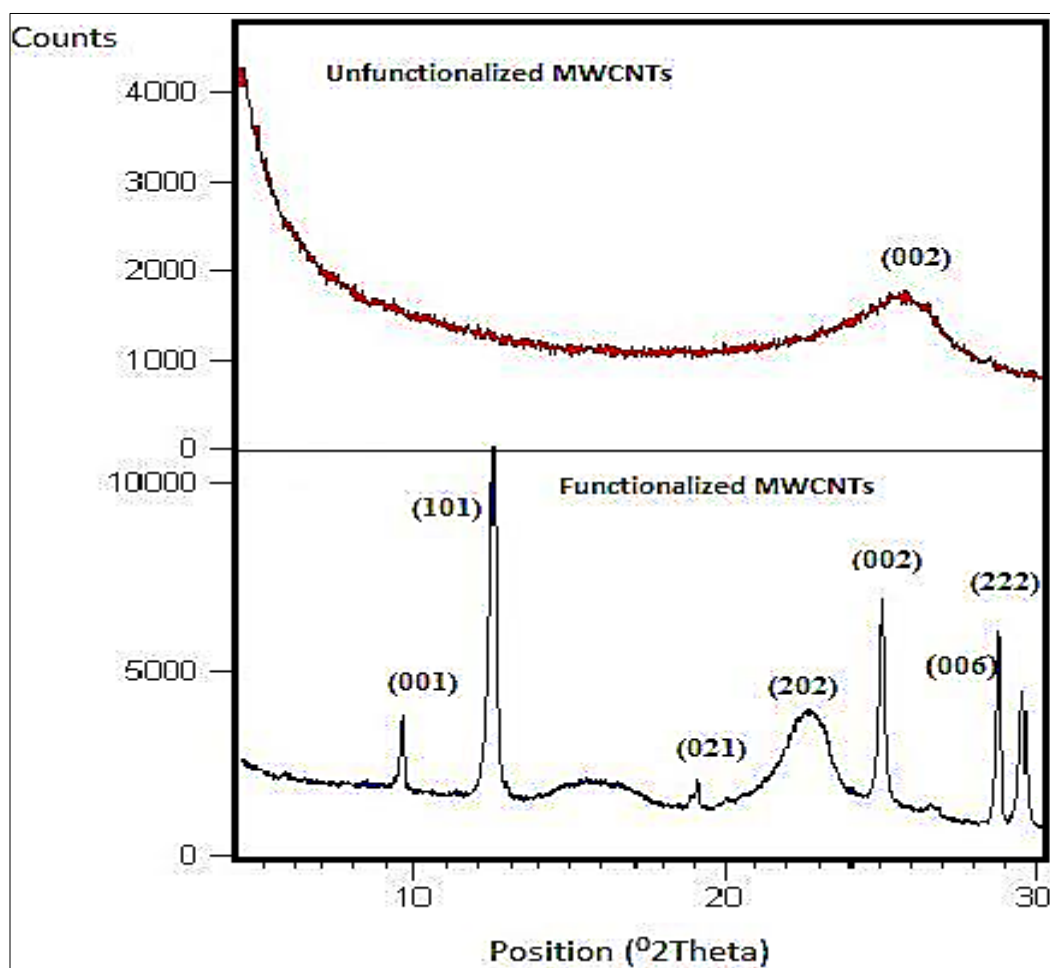


Fig 2: XRD spectrum of unfunctionalized and functionalized MWCNTs

Dried and functionalized MWCNTs were poured in a mixture of 2 mL N, N-dimethylformamide and 0.2 mL Nafion {a conducting polymer (0.5% in ethanol)}. Now, solution was made up to 30 mL with double distilled water followed by hot sonication for 3 hr to get homogeneous suspension.

2.3.5 Modification of GCE

The optimised volume of 15 μ L of f-MWCNT suspension was placed on the surface of pretreated GCE with the help of a micropipette and allowed to dry at room temperature.

3. Results and Discussion

3.1 Surface characterization of f-MWCNT-Nafion/ GCE

Morphology of bare and f-MWCNT-Nafion modified GCE was studied by taking SEM (scanning electron microscope) images of both bare GCE (Figure 3A) and f-MWCNT-Nafion modified GCE (Figure 3B) using a ZEISS-EV018 instrument from University Science and Instrumentation Centre (University of Rajasthan, Jaipur). Modification of bare GCE with f-MWCNT-Nafion significantly increased peak responses which were attributed to the increased

surface area and increased electron transfer rate between

electrode and electrolyte interface due to modification.

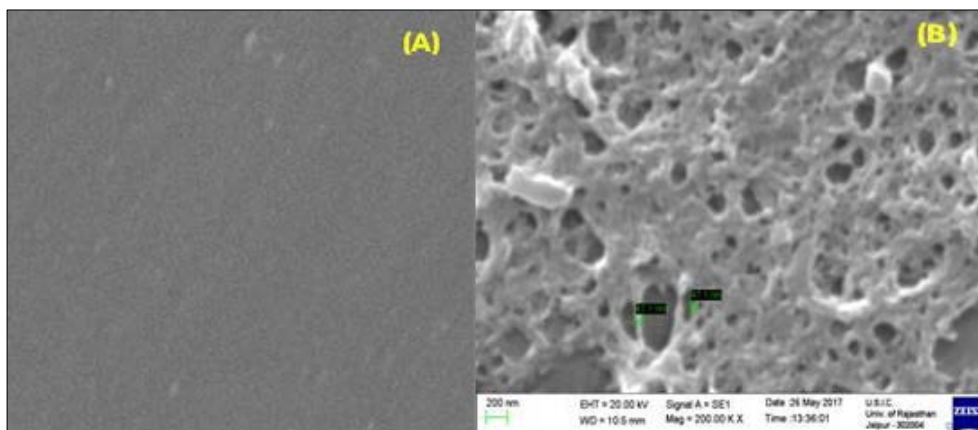


Fig 3: SEM images of (A) bare GCE (B) f-MWCNT-Nafion modified surface of GCE (two dark spots in Figure B show observed diameter of nanotubes of 47.1nm)

3.2 Electrochemical behaviour of melphalan

Voltammetric techniques (CV, DPV), controlled potential coulometry (CPC) and chronocoulometry (CC) were performed to investigate electrochemical behaviour of melphalan. Redox kinetics and quantitative determination were studied at both bare and f-MWCNT-Nafion/GCE. Results thus obtained were compared to establish f-MWCNT-Nafion/GCE as a sensor for better characterization and determination of melphalan.

3.2.1 Cyclic voltammetric behaviour

Cyclic voltammogram was recorded in the potential range of 0.0 V to 1.1 V vs. Ag/AgCl reference electrode in BR buffer of pH 7.2 at 100 mV s⁻¹ scan rate. A well-defined oxidation peak was obtained at 0.866 V and 0.821 V using bare GCE and f-MWCNT-Nafion/GCE respectively in anodic direction but there was no corresponding cathodic peak which indicated about irreversible nature of oxidation process.

3.2.2 Effect of scan rate

Effect of scan rate on peak current (I_p) and peak potential (E_p) was examined under above mentioned conditions. As the scan rate was increased from 10 to 200 mV s⁻¹ for both bare GCE (Figure 4) and f-MWCNT-Nafion/GCE (Figure 5) at 2.27×10^{-4} M concentration of melphalan, peak potential shifted towards more positive values with increase in peak current confirming the irreversible nature of the oxidation process [17, 18].

A linear Randles-Sevcik plot (I_p vs. $v^{1/2}$) was obtained for both bare GCE and f-MWCNT-Nafion/GCE (Inset a of Figure 4 and 5 respectively), which indicated about diffusion controlled mass transport in rate determining step of oxidation process [19, 20]. Corresponding linear regression equations were expressed as:

$$I_p (\mu A) = 3.2805 v^{1/2} (mV s^{-1})^{1/2} - 3.1172 \quad (R^2 = 0.9802) \quad \text{(Bare GCE)} \quad (1)$$

$$I_p (\mu A) = 8.6402 v^{1/2} (mV s^{-1})^{1/2} - 27.021 \quad (R^2 = 0.9239) \quad \text{(f-MWCNT-Nafion/GCE)} \quad (2)$$

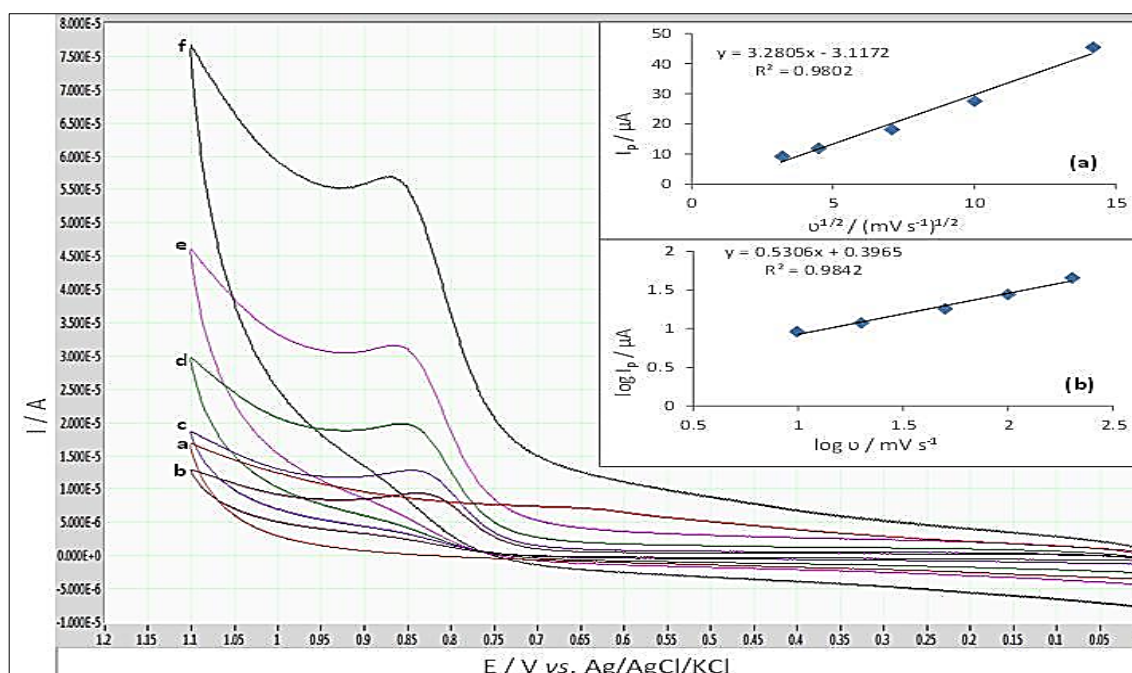


Fig 4: Cyclic Voltammograms of melphalan (2.27×10^{-4} M) at bare GCE at different scan rates (a) blank, (b) 10 mV s⁻¹, (c) 20 mV s⁻¹, (d) 50 mV s⁻¹, (e) 100 mV s⁻¹ and (f) 200 mV s⁻¹ in BR Buffer (pH 7.2); Inset (a) Plot of I_p vs. $v^{1/2}$; Inset (b) Dependence of $\log I_p$ vs. $\log v$

To confirm above findings, a graph between logarithm of peak current ($\log I_p$) and logarithm of scan rate ($\log v$) was plotted for bare and modified GCE (Inset b of Figure 4 and 5 respectively). Corresponding linear regression equations

were expressed as:

$$\log I_p (\mu A) = 0.5306 \log v (mV s^{-1}) + 0.3965 (R^2 = 0.9842) \text{ (Bare GCE) (3)}$$

$$\log I_p (\mu A) = 0.5513 \log v (mV s^{-1}) + 0.1238 (R^2 = 0.9744) \text{ (f-MWCNT-Nafion/GCE) (4)}$$

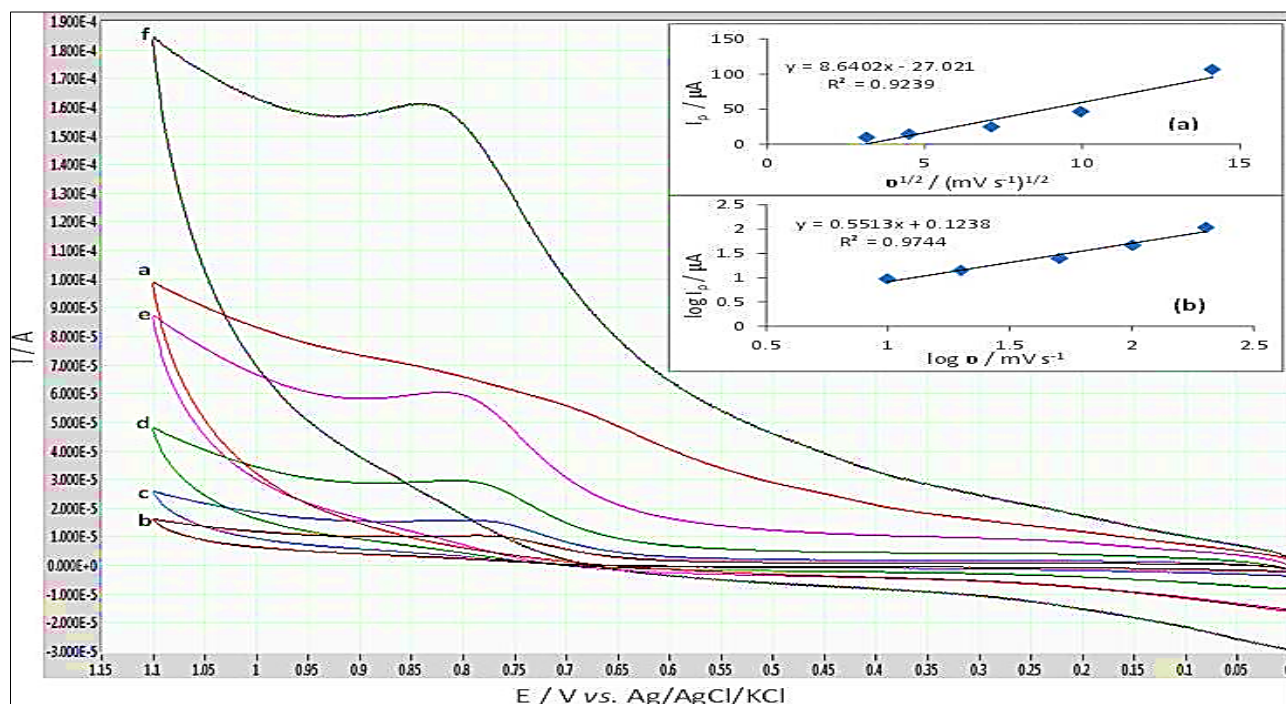


Fig 5: Cyclic Voltammograms of melphalan (2.27×10^{-4} M) at f-MWCNT-Nafion/GCE at different scan rates (a) blank, (b) 10 mV s^{-1} , (c) 20 mV s^{-1} , (d) 50 mV s^{-1} , (e) 100 mV s^{-1} and (f) 200 mV s^{-1} in BR Buffer (pH 7.2); Inset (a) Plot of I_p vs. $v^{1/2}$; Inset (b) Dependence of $\log I_p$ vs. $\log v$

Slope values of $\log I_p$ vs. $\log v$ plot for bare and modified electrodes were 0.5306 and 0.5513 respectively (eq. 3 and 4 respectively) which were in close agreement with theoretical slope value of 0.5 for pure diffusion controlled process [21].

3.2.3 Determination of surface area of bare GCE and MWCNT-Nafion/GCE: To determine effective surface area of electrodes, cyclic voltammograms of $1 \text{ mM K}_3[\text{Fe}(\text{CN})_6]$ solution were recorded at different scan rates using both bare GCE and f-MWCNT-Nafion/GCE. KCl solution of 0.1 M concentration was used as the supporting electrolyte. Peak potentials for reversible redox couple $[\text{Fe}(\text{CN})_6]^{3-}/[\text{Fe}(\text{CN})_6]^{4-}$ were $0.156/0.283 \text{ V}$ and $0.151/0.275 \text{ V}$ at bare GCE and f-MWCNT-Nafion/GCE respectively at 100 mV s^{-1} scan rate. Now, a plot of I_p vs. $v^{1/2}$ was obtained according to the following equation:

$$I_p = 0.4463 \left(\frac{F^3}{RT} \right)^{1/2} A n^{3/2} D_o^{1/2} C_o v^{1/2} \quad (5)$$

Where, I_p is peak current in Ampere, A is electrode surface area in cm^2 , F is Faraday's constant ($96,485 \text{ C/mol}$) and R is the universal gas constant (8.314 J/mol K). For $1 \text{ mM K}_3[\text{Fe}(\text{CN})_6]$ diffusion coefficient (D_o) = $7.6 \times 10^{-6} \text{ cm}^2/\text{s}$ and total number of electrons (n) equal = 1, C is concentration in M and v is the scan rate in V s^{-1} . Effective surface areas of bare GCE and f-MWCNT-Nafion/GCE were calculated from the slope of I_p vs. $v^{1/2}$ plot and were found as 0.0541 cm^2 and 0.0982 cm^2 respectively.[22] Thus, modification increased surface area of bare electrode by 0.0441 cm^2 which is 81.52% of bare one resulting in improved peak response towards the oxidation of drug moiety.

3.2.4 Oxidation kinetics of melphalan Hydrochloride

3.2.4.1 Determination of parameter $[\alpha n']$

To calculate value of $\alpha n'$, for diffusion controlled process, E_p vs. $\log v$ graph was plotted for both bare GCE (Figure 6A) and modified GCE (Figure 6B).

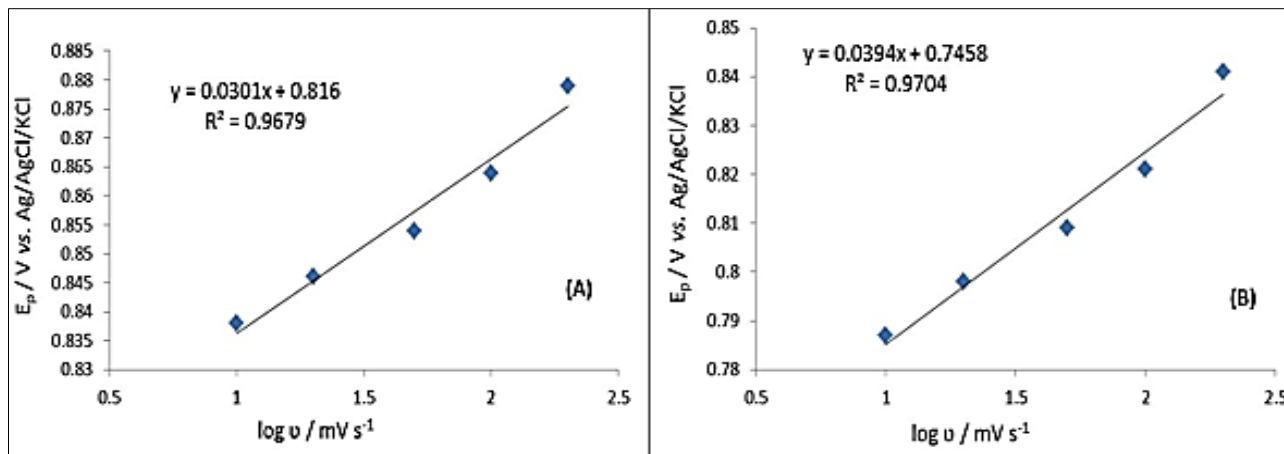


Fig 6: Variation of peak potential (E_p) versus logarithm of scan rate ($\log v$) at (A) Bare GCE and (B) f-MWCNT-Nafion/GCE for melphalan ($2.27 \times 10^{-4} \text{ M}$) in BR Buffer (pH 7.2)

According to the following equation given by Nicholson and Shain:^[23, 24]

$$E_p = E^0 + \frac{RT}{\alpha n' F} \left[0.78 + \ln \left(\frac{D_0^{1/2}}{k_s} \right) - 0.5 \ln \frac{RT}{\alpha n' F} \right] + \left(\frac{RT}{\alpha n' F} \right) \ln v \quad (6)$$

Where E_p is peak potential (V), E^0 is formal potential (V), α is the anodic electron transfer coefficient, n' is number of electrons involved in slowest step, k_s is electrochemical heterogeneous rate constant, D_0 is diffusion coefficient (cm^2/s) and rest parameters have their usual meanings. Now, value of $\alpha n'$ was calculated by comparing slope value from Figure 6A and Figure 6B with eq. (6) which came out equal to 0.982 and 0.75 for bare GCE and modified GCE respectively.

3.2.4.2 Determination of total number of electrons

Controlled potential coulometry (CPC) technique was performed to determine total number of electrons (n) involved in oxidation process. Number of Transferred

electrons (n) was calculated from charge consumed by desired concentration of Melphalan hydrochloride. For this purpose 5 ml of 4 mg/ml solution of melphalan hydrochloride was taken in an electrochemical cell and electrolyzed at constant potential of 0.867 V vs. Ag/AgCl reference electrode. During electrolysis, solution was kept stirred and a continuous stream of instrumental grade nitrogen gas was allowed to pass through it to get oxygen free environment. Using equation $Q = nFN$, (where, Q is charge in coulombs, F is Faraday's constant and N is number of moles of substrate) total number of electrons participated (n) was calculated as 2^[25].

3.2.4.3 Determination of diffusion coefficient (D_0)

Double potential step chronocoulometry technique was performed to determine the value of diffusion coefficient using both bare and modified electrode. In this technique charge (Q) vs. square root of time ($t^{1/2}$) plot (Anson plot) was recorded which was linear for both bare GCE (Figure 7A) and modified GCE (Figure 7B).

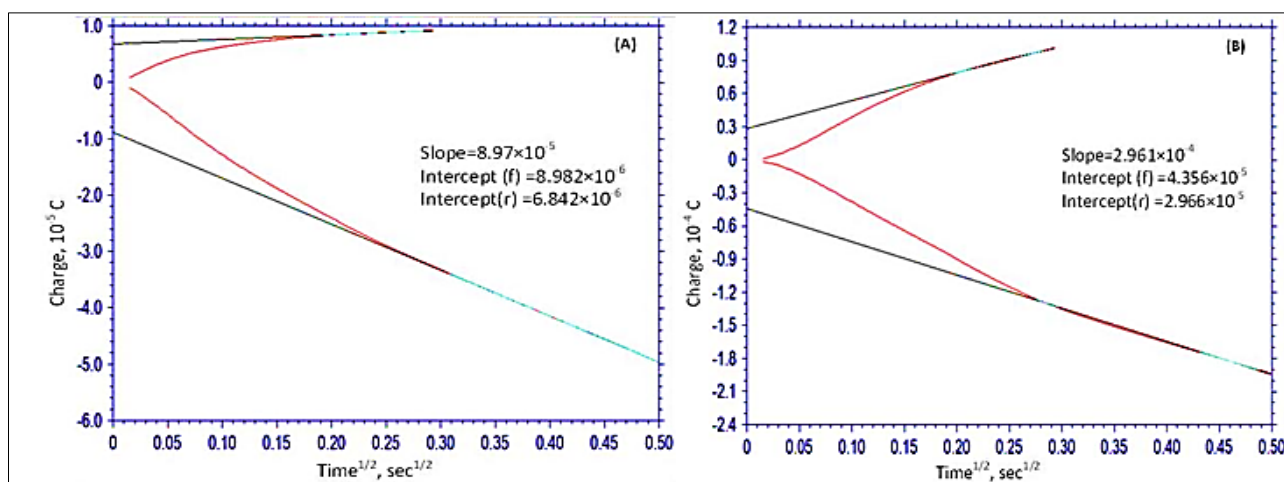


Fig 7: Anson plot (Q vs. $t^{1/2}$) for oxidation of melphalan ($2.27 \times 10^{-4} \text{ M}$) from two step Chronocoulometry at (A) bare GCE and (B) f-MWCNT-Nafion/GCE in BR Buffer (pH 7.2)

Diffusion coefficient (D_0) was calculated by comparing slope value, obtained from Anson plot, with corresponding parameters of Anson equation (eq. 7).^[26] Diffusion coefficients were found to be $1.12 \times 10^{-3} \text{ cm}^2/\text{s}$ and $3.722 \times 10^{-3} \text{ cm}^2 \text{ s}^{-1}$ for bare GCE and modified GCE respectively.

$$Q_d = \frac{2nFAC_0 D_0^{1/2} t^{1/2}}{\pi^{1/2}} \quad (7)$$

Where Q_d is charge due to electrolysis of diffusing species (coulomb), n is number of electrons transferred per ion or

molecule, D_0 is diffusion coefficient ($\text{cm}^2 \text{s}^{-1}$) and t is time (sec) rest parameters have their usual meanings.

3.2.4.4 Determination of heterogeneous electrochemical rate constant (k_s)

Rate constant for oxidation process of melphalan was

calculated from the intercept of E_p vs. $\log v$ plot from equation (6) given by Nicholson and Sain if the value of E^0 was known. To calculate the value of E^0 , line of E_p vs. v plot (Figure 8A and Figure 8B for bare GCE and modified GCE respectively) was extrapolated till v falls to 0 and the intercept value, thus obtained, was used as E^0 [27-29].

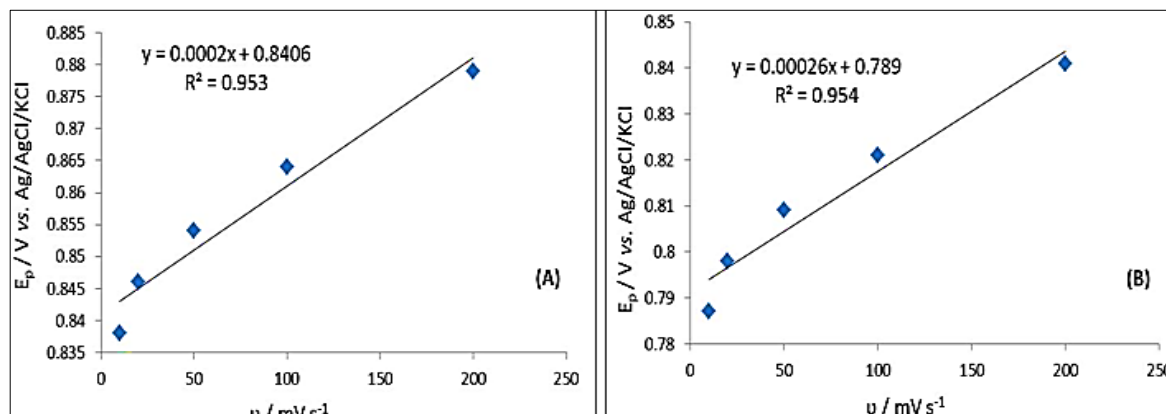


Fig 8: Plot of peak potential (E_p) versus scan rate (v) for melphalan ($2.27 \times 10^{-4} \text{M}$) at (A) bare GCE and (B) f-MWCNT-Nafion/GCE in BR Buffer (pH 7.2)

Calculated value of heterogeneous rate constant (k_s) was 1.18 s^{-1} and 2.60 s^{-1} with E^0 values of 0.841 V and 0.789 V for bare GCE and f-MWCNT-Nafion/GCE respectively

Table 1: Comparison of results obtained from bare GCE and modified sensor

Parameters	Bare GCE	MWCNT-Nafion/GCE
D_0 ($\text{cm}^2 \text{s}^{-1}$)	1.12×10^{-3}	3.722×10^{-3}
k_s (s^{-1})	1.18	2.60

Increased values of rate constant and diffusion coefficient prove that modification catalyses electron transfer and oxidation process by providing large electroactive surface area. Comparative data were shown in Table 1.

3.3 Effect of pH

Effect of pH on electrochemical response of melphalan hydrochloride, with effective concentration of $2.27 \times 10^{-4} \text{ M}$, was studied in the pH range of 2-12 of BR buffer using DPV technique. Sharp oxidation peaks were obtained only in the

pH range of 2.4-8.4 (Figure 9) and after that sharp oxidation peaks were disappeared. As the pH was increased peak current (I_p) and sharpness were firstly increased, attained maxima at pH 7.2 and then decreased (Inset a of Figure 9) hence on the basis of peak height (I_p) and peak sharpness pH value of 7.2 was selected for the whole voltammetric studies. In the pH range of 2.4-8.4, peak potential (E_p) vs. pH plot was nearly linear which clearly indicated about the participation of proton in the oxidation mechanism of melphalan (Inset b of Figure 9) [30].

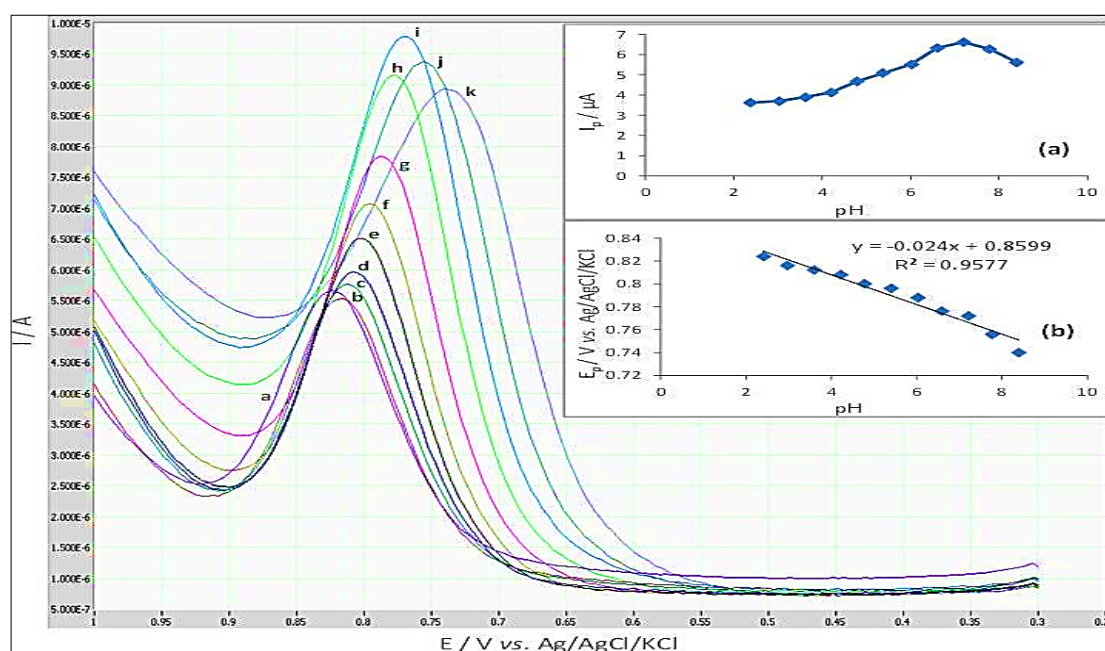


Fig 9: Differential pulse voltammograms of melphalan ($2.27 \times 10^{-4} \text{M}$) with varying pH (a) pH 2.4 (b) pH 3.0 (c) pH 3.6 (d) pH 4.2 (e) pH 4.8 (f) pH 5.4 (g) pH 6 (h) pH 6.6 (i) pH 7.2 (j) pH 7.8 (k) pH 8.4; Inset (a) Plot of I_p vs. pH; Inset (b) Dependence of E_p vs. pH

Linearity of E_p vs. pH plot was expressed by the following linear regression equation:

$$E_p \text{ versus Ag/AgCl (V)} = -0.024\text{pH} + 0.859 \quad (R^2 = 0.9577) \quad (8)$$

Furthermore, slope of E_p vs. pH plot should be equal to $2.303\delta RT/nF$ according to following the equation:

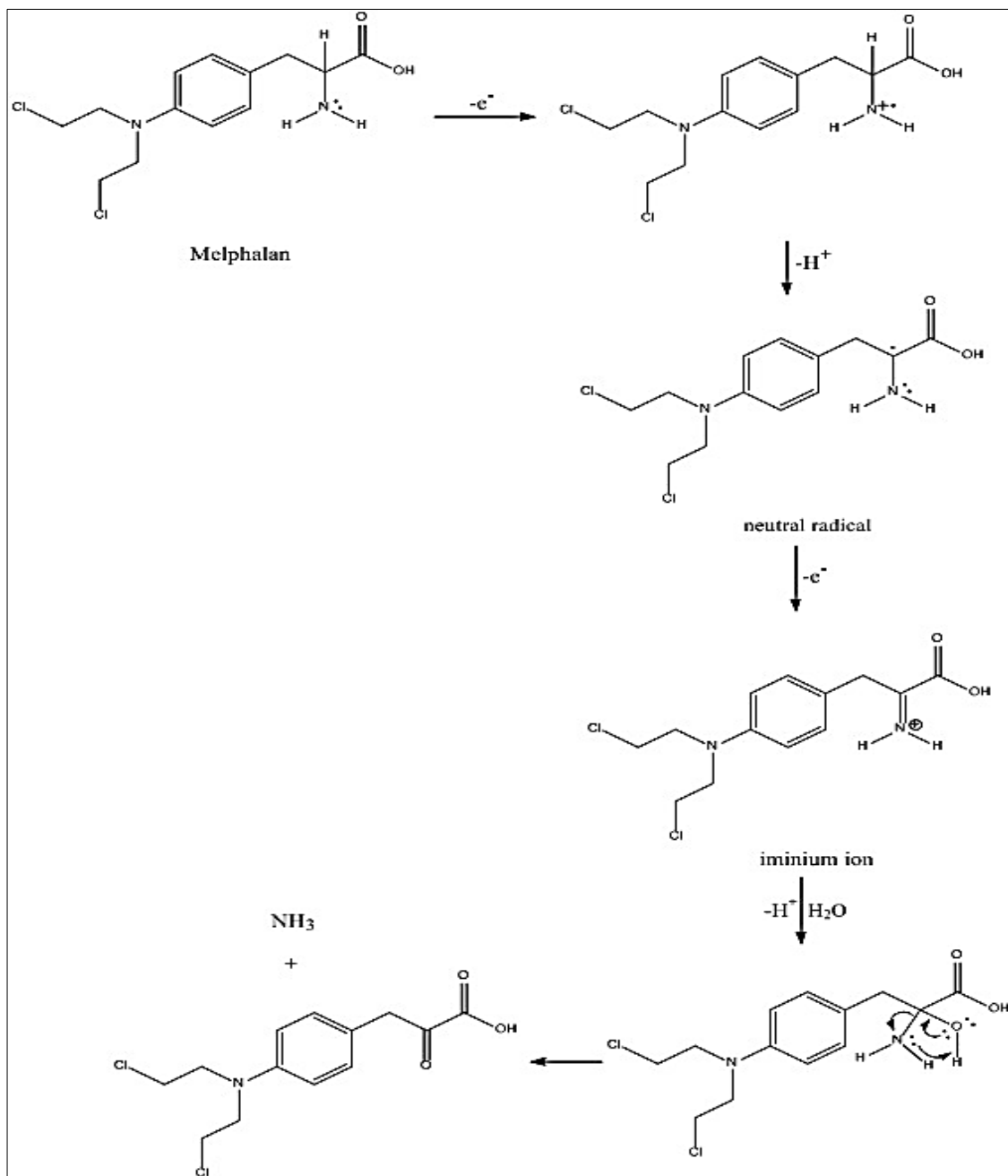
$$E_p \text{ versus Ag/AgCl (V)} = E^0 + \frac{RT}{nF} \ln \left[\frac{O}{R} \right] - \delta \frac{RT}{nF} \ln [H^+] \quad (9)$$

Where, δ = number of protons and rest parameters have their usual meanings. Slope of E_p vs. pH plot was 24

mV/pH, which is very far from theoretical value of 59 mV/pH for $1H^+/1e^-$ participation [31]. Furthermore, comparison of slope of equations (8) and (9) gave the value of δ/n equals to 0.41 which may be taken as 0.5. Since, CPC indicated participation of $2e^-$ in oxidation so it might be concluded as $1H^+/2e^-$ oxidation process.

3.4 Proposed oxidation mechanism

Based on CV, CPC and pH studies, Scheme 1 was proposed for the oxidation of melphalan where removal of one proton and two electrons take place in electrochemical step. Literature also supports such mechanism [32, 34].



Scheme 1: Reaction mechanism for oxidation of melphalan.

3.5 Electroanalytical determination of melphalan

The electroanalytical techniques are quite effective and rapid for analytical purposes. Especially, DPV technique is quite sensitive to the measurement of very low amount of

numerous organic compounds. The developed f-MWCNT-Nafion/GCE sensor showed a greater peak current in comparison to the bare GCE (Figure 10).

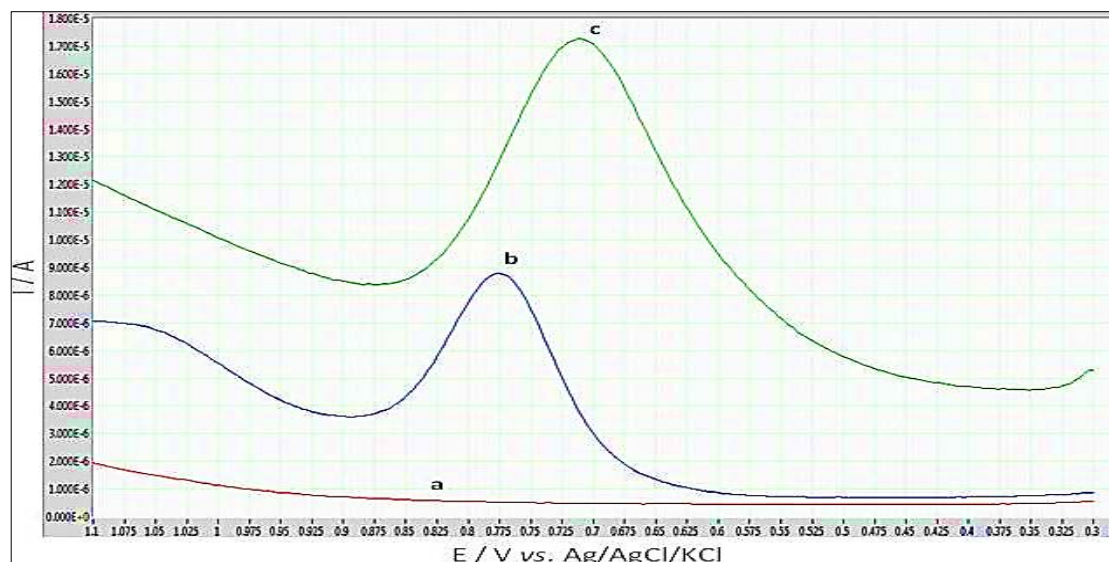


Fig 10: Comparative differential pulse voltammograms for melphalan (a) blank (b) bare GCE (c) f-MWCNT-Nafion/GCE in BR Buffer (pH 7.2)

So, in the present paper DPV was optimized for determination of melphalan in pharmaceutical formulations, bulk form and serum as biological sample.

3.5.1 Optimisation of operational parameters

Differential pulse voltammetric response depends on various

operational parameters such as scan increment (ΔS), pulse width, pulse period, pulse amplitude etc. So prior to record the differential pulse voltammograms of melphalan solutions aforementioned parameters were optimized and are tabulated in Table 2.

Table 2: Optimized operational parameters for determination of melphalan using DPV technique

DPV	
Scan increment (mV)	04
Pulse amplitude (mV)	50
Pulse width (s)	0.02
Pulse period (s)	0.5

Volume of f-MWCNT-Nafion suspension, to be drop down on GCE, was also optimized on the basis of peak shape and peak height of differential pulse voltammograms. Volume of 15 μL of f-MWCNT-Nafion suspension was optimised to modify bare GCE.

melphalan on peak current (I_p) was studied in the working concentration range of 9.99×10^{-10} M to 5.19×10^{-7} M using DPV method at f-MWCNT-Nafion/GCE. Peak current (I_p) was found to be increased linearly with melphalan concentration over the range of 1.7×10^{-8} to 3.9×10^{-7} M (Figure 11).

3.5.2 Effect of Concentration: Effect of concentration of

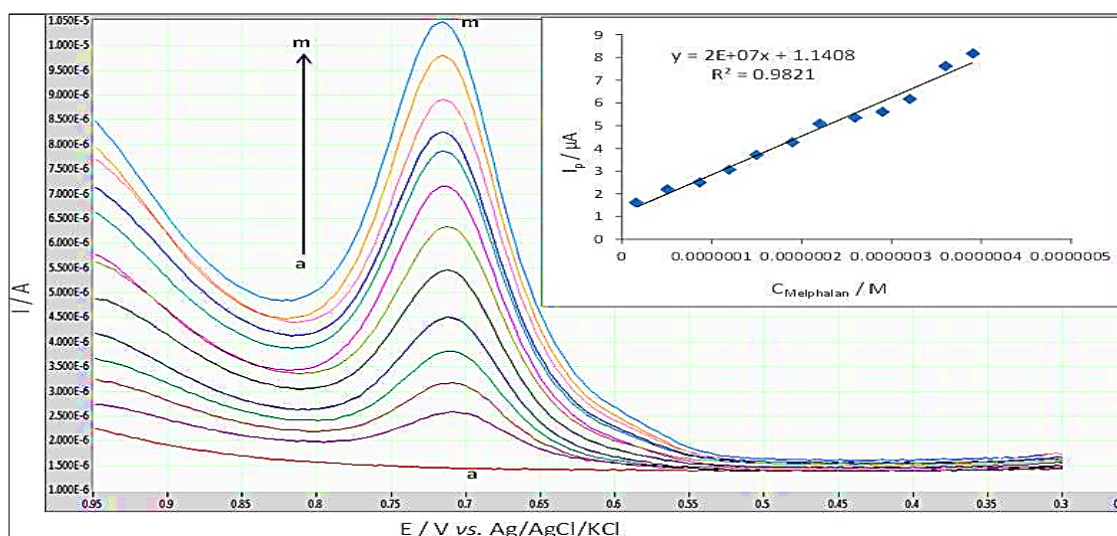


Fig 11: The Differential pulse Voltammograms of melphalan at MWCN-Nafion/GCE with varying concentrations (a) blank, (b) 1.7×10^{-8} M, (c) 5.2×10^{-8} M, (d) 8.6×10^{-8} M, (e) 1.2×10^{-7} M, (f) 1.5×10^{-7} M, (g) 1.9×10^{-7} M, (h) 2.2×10^{-7} M, (i) 2.6×10^{-7} M, (j) 2.9×10^{-7} M, (k) 3.2×10^{-7} M, (l) 3.6×10^{-7} M and (m) 3.9×10^{-7} M in BR buffer (pH 7.2); (Inset) Plot of I_p vs. melphalan concentration

For quantification of melphalan calibration curve was obtained by plotting peak current (I_p) against melphalan concentration (Inset of Figure 6) which was expressed by the following linear regression equation:

$$I_p(\mu A) = 2 \times 10^7 C_{\text{Melphalan}} (M) + 1.1408 (R^2 = 0.9821) \quad (10)$$

3.5.3 Validation of the developed analytical method

To validate the proposed method, for quantification of melphalan, various validation characteristics were evaluated such as limit of detection (LOD), limit of quantification (LOQ), accuracy, precision (in terms of repeatability and

reproducibility) and percentage recovery [35, 38].

3.5.3.1 Detection and quantification limit

The limit of detection (LOD) and limit of quantification (LOQ) were calculated using the following relationships:

$$LOD = \frac{3s}{m}, LOQ = \frac{10s}{m}$$

Where, s and m are standard deviation of intercept and slope respectively of peak current vs. concentration plot. Analytical parameters, LOD, LOQ and % RSD are listed in Table 3.

Table 3: Analytical parameters for voltammetric determination of melphalan in bulk form using DPV at MWCNT-Nafion/GCE

Techniques	DPV
Measure potential (V)	0.715
Linearity range (M)	1.70×10^{-8} – 3.9×10^{-7}
Slope ($\mu A/M$)	2.0×10^7
Intercept (μA)	1.1408
Correlation coefficient	0.9821
LOD (M)	1.95×10^{-9}
LOQ (M)	1.30×10^{-8}
SD	0.013
% RSD	1.14

LOD of the developed method was also compared with other reported methods in literature as shown in Table 4 [6].

Table 4: Comparison of different methods used for the determination of melphalan

Analytical method	LOD	References
DPV	1.0×10^{-6} M	[12]
GC-MS	-	[11]
HPLC	10 ng/mL	[39]
LC	5 ng/mL	[40]
RP-HPLC	2 ng/mL	[41]
HPLC	-	[42]
Ion-pair HPLC	5 ng/ml	[7]
RP-HPLC	50 ng/ml	[43]
Calorimetric method	15.7 ng/ml	[44]
Thin chromatography	20 ng/ml	[45]
Electrochemical MIP sensor	5.40 ng/ml	[6]
This work	0.67 ng/ml (1.95×10^{-9} M)	This work

Low values of LOD and LOQ showed good sensitivity and low value of % RSD indicated about good precision of the developed method

3.5.3.2 Accuracy and % recovery

To determine the accuracy and % recovery of the developed method a certain amount of melphalan sample was added in an appropriate amount of BR buffer solution. Now, the concentration of melphalan was determined using standard

addition method. The error between amount added and amount found was expressed in terms of mean accuracy (n=5) whereas mean % recovery was expressed in terms of mean $\pm$ SD and was found equal to 99.96 ± 0.0027 . Accuracy and % recovery data was abridged in Table 5.

Table 5: Recovery results for voltammetric determination of bulk melphalan using DPV at f-MWCNT-Nafion/GCE

Amount added ($\mu g/L$)	Mean amount found* ($\mu g/L$)	Mean Accuracy/ Error*	Mean % recovery*	SD	% RSD
10.0	9.996	0.004	99.86	0.0026	0.026
20.0	19.995	0.005	99.92	0.0012	0.006
30.0	30.009	0.009	100.1	0.0042	0.014
Mean $\pm$ SD = 99.96 ± 0.0027					
*mean of five replicate readings (n=5)					

Results confirmed good accuracy and % recovery of the proposed method.

3.5.3.3 Precision

To estimate the precision of the proposed method intraday precision (repeatability) and interday precision (reproducibility) were calculated using DPV technique. For intraday precision three test melphalan solutions of concentration 10 $\mu g/L$, 20 $\mu g/L$ and 30 $\mu g/L$ were analysed

for three times (n=3) at an equal time interval of 0 hr, 2 hr and 4 hr whereas for interday precision aforementioned three test melphalan solutions were analysed for three times (n=3) at an equal time interval of 24 hr for three days. Results of intraday precision and interday precision were abridged in Table 6 and Table 7 respectively.

Table 6: Precision of the proposed method in terms of repeatability (intraday)

Conc. added ($\mu\text{g/L}$)	Peak current (μA)			Conc. Found ($\mu\text{g/L}$)			Mean $\pm$ SD	% RSD
	0 hr	2 hr	4 hr	0 hr	2 hr	4 hr		
10.0	1.994	1.982	1.975	9.998	9.982	9.969	9.983 $\pm$ 0.0032	0.032
20.0	2.288	2.275	2.262	19.996	19.982	19.965	19.981 $\pm$ 0.0024	0.012
30.0	2.642	2.635	2.621	30.009	29.992	29.979	29.993 $\pm$ 0.0010	0.0033
Mean% RSD = 0.016								
*mean of three replicate readings (n=3)								

Table 7: Precision of the proposed method in terms of reproducibility (interday)

Conc. added ($\mu\text{g/L}$)	Concentration found ($\mu\text{g/L}$)			Mean $\pm$ SD	% RSD
	D1*	D2*	D3*		
10.0	9.986	9.978	9.959	9.974 $\pm$ 0.028	0.28
20.0	20.004	19.985	19.961	19.983 $\pm$ 0.021	0.11
30.0	30.002	29.992	29.985	29.993 $\pm$ 0.011	0.037
Mean% RSD = 0.14					
*mean of three replicate readings (n=3)					

Low values of % RSD as 0.016 and 0.14 for repeatability and reproducibility respectively confirmed good precision of the developed procedure for the determination of melphalan.

3.5.4 Interference studies: The tolerance limit was defined as maximum concentration of interferent that causes an error

of less than $\pm 5\%$ for determination of melphalan. For interference studies effect of some commonly used interferents in pharmaceutical preparations was evaluated on DPV response of 2.27×10^{-4} M melphalan solution as a standard.

Table 8: Effect of excipients on assay of melphalan

Interferent	Concentration (M)	Signal change (%)
Starch	0.02	+1.76
D-glucose	0.02	+1.51
sucrose	0.02	+1.80
Dextrose	0.02	+0.98
NaCl	0.02	-1.12

Obtained results (Table 8) showed that 100-fold excess concentration of starch, D-glucose, sucrose, dextrose and NaCl did not interfere with the DPV signal of melphalan which justified that the developed method can be considered as specific.

3.5.5 Determination of melphalan in pharmaceutical formulation and spiked serum sample: To validate the proposed method for practical applications, quantification of

melphalan was carried out in pharmaceutical formulation and spiked serum sample using standard addition method. Procedure for pharmaceutical and spiked serum analysis was adopted from sections pharmaceutical sample preparation and spiked serum sample preparation respectively. Obtained results for spiked serum sample were listed in Table 9 and Table 10 while results of pharmaceutical formulation were tabulated in Table 11.

Table 9: Recovery results of the proposed method for spiked serum samples (solution of standard melphalan was spiked)

Sample	Amount added ($\mu\text{g/L}$)	Amount found ($\mu\text{g/L}$)	% recovery*	SD	% RSD
Standard in serum sample	30.0	30.02, 30.03, 29.95, 29.98, 29.98	29.994 $\pm$ 0.045 $t_{\text{cal}} = 0.37$ $t_{\text{tab}} = 2.78$	0.036	0.12

*Results of recovery values are given as mean $\pm$ ts/ $\sqrt{n}$ (at 95% confidence level)

Table 10: Recovery results of the proposed method for spiked serum samples (solution of pharmaceutical formulation was spiked)

Sample	Amount added ($\mu\text{g/L}$)	Amount found ($\mu\text{g/L}$)	% recovery*	SD	% RSD
Melphalan in serum sample	30.0	30.09, 30.01, 29.91, 29.88, 29.85	29.948 $\pm$ 0.12 $t_{\text{cal}} = 1.73$ $t_{\text{tab}} = 2.78$	0.099	0.33

*Results of recovery values are given as mean $\pm$ ts/ $\sqrt{n}$ (at 95% confidence level)

Table 11: Estimation of melphalan in drug (MEGVAL-50mg)

Labelled quantity in vial (mg/vial)	Quantity found in vial (mg/vial)	% recovery*	SD	% RSD
50	49.198, 49.191, 50.02, 50.06, 49.192	49.53 $\pm$ 0.57 $t_{\text{cal}} = 2.28$ $t_{\text{tab}} = 2.78$	0.46	0.93

*Results of recovery values are given as mean $\pm$ ts/ $\sqrt{n}$ (at 95% confidence level)

Good recovery of melphalan indicated that proposed method can be successfully applied for direct determination in pharmaceutical formulation and spiked serum sample.

4. Conclusions

In this work, detailed oxidation behaviour of melphalan, including determination of different kinetic parameters, was studied at both bare GCE and f-MWCNT-Nafion modified GCE in BR buffer at pH 7.2. Comparative studies at both bare GCE and f-MWCNT-Nafion/GCE clearly showed that f-MWCNT-Nafion positively catalyzed the electron transfer rate between electrode and electrolyte interface in the oxidation process of melphalan. Improved electron transfer increased rate constant and peak current of voltammetric response whereas decreased peak potential that is required for oxidation process. Oxidation process was diffusion controlled and involved two electrons and one proton. A possible oxidation mechanism of melphalan was also proposed based on the experimental observations. Furthermore, a fast, cost effective and reliable differential pulse voltammetric (DPV) method was developed using f-MWCNT-Nafion/GCE based sensor and successfully applied for the determination of melphalan in bulk formulation, pharmaceutical formulation and serum as a biological sample. Good sensitivity, selectivity as well as good accuracy and precision validated the developed method for the determination of melphalan. Already reported methods often require pre-treatment or derivatization of melphalan to remove matrix effect and were not economical as well as time effective. But owing to simple and economical nature the developed method can be a good alternative to the already proposed methods in literature for quality control laboratories and pharmacokinetic studies.

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